

Overview of Mean squared error

Subject: Mean squared error

$$\text{MSE} = \frac{1}{n} \sum_{i=1}^n (y_i - \hat{y}_i)^2$$

1.

In regression analysis, plotting is a more natural way to view the overall trend of the whole data. The mean of the distance from each point to the predicted regression model can be calculated, and shown as the mean squared error. The squaring is critical to reduce the complexity with negative signs. To minimize MSE, the model could be more accurate, which would mean the model is closer to actual data. One example of a linear regression using this method is the least squares method—which evaluates appropriateness of linear regression model to model bivariate dataset, but whose limitation is related to known distribution of the data. The term mean squared error is sometimes used to refer to the unbiased estimate of error variance: the residual sum of squares divided by the number of degrees of freedom. This definition for a known, computed quantity differs from the above definition for the computed MSE of a predictor, in that a different denominator is used. The denominator is the sample size reduced by the number of model parameters estimated from the same data, $(n-p)$ for p regressors or $(n-p-1)$ if an intercept is used (see errors and residuals in statistics for more details). Although the MSE (as defined in this article) is not an unbiased estimator of the error variance, it is consistent, given the consistency of the predictor. In regression analysis, "mean squared error", often referred to as mean squared prediction error or "out-of-sample mean squared error", can also refer to the mean value of the squared deviations of the predictions from the true values, over an out-of-sample test space, generated by a model estimated over a particular sample space. This also is a known, computed quantity, and it varies by sample and by out-of-sample test space. In the context of gradient descent algorithms, it is common to introduce a factor of $1/2$ to the MSE for ease of computation after taking the derivative. So a value which is technically half the mean of squared errors may be called the MSE.

2.

Applications Minimizing MSE is a key criterion in selecting estimators; see minimum mean-square error. Among unbiased estimators, minimizing the MSE is equivalent to minimizing the variance, and the estimator that does this is the minimum variance unbiased estimator. However, a biased estimator may have lower MSE; see estimator bias. In statistical modelling the MSE can represent the difference between the actual observations and the observation values predicted by the model. In this context, it is used to determine the extent to which the model fits the data as well as whether removing some explanatory variables is possible without significantly harming the model's predictive ability. In forecasting and prediction, the Brier score is a measure of forecast skill based on MSE.

3.

$$\begin{aligned} \text{MSE}(\hat{S}_a^2) &= E[(n-1)a^2 S_{n-1}^2 - \sigma^2]^2 = E[(n-1)^2 a^4 S_{n-1}^4 - 2(n-1)a^2 S_{n-1}^2 \sigma^2 + \sigma^4] \\ &= (n-1)^2 a^4 E[S_{n-1}^4] - 2(n-1)a^2 \sigma^2 E[S_{n-1}^2] + \sigma^4 = (n-1)^2 a^4 E[S_{n-1}^4] - 2(n-1)a^2 \sigma^4 + \sigma^4 E[S_{n-1}^2] \\ &= \sigma^2 = (n-1)^2 a^2 (\gamma^2 n + n + 1) - 2(n-1)a^2 \sigma^4 + \sigma^4 E[S_{n-1}^2] = \text{MSE}(S_{n-1}^2) + \sigma^4 = n-1 na^2 ((n-1)\gamma^2 + n^2 + n) \sigma^4 - 2(n-1)a^2 \sigma^4 + \sigma^4 \end{aligned}$$

4.

$$\text{MSE}(\bar{X}) = E[(\bar{X} - \mu)^2] = (\sigma^2/n) = \sigma^2/n \quad \left\{ \text{displaystyle } \operatornamename {MSE} \left(\overline{X} \right) = E \left[\left(\overline{X} - \mu \right)^2 \right] = \left(\frac{\sigma^2}{n} \right) \right.$$

5.

In other words, the MSE is the mean $\left(\frac{1}{n} \sum_{i=1}^n \right)$ of the squares of the errors $(Y_i - \hat{Y}_i)^2$. This is an easily computable quantity for a particular sample (and hence is sample-dependent). In matrix notation,

6.

$$S_{n-1}^2 = \frac{1}{n-1} \sum_{i=1}^n (X_i - \bar{X})^2 = \frac{1}{n-1} \left(\sum_{i=1}^n X_i^2 - n\bar{X}^2 \right) \quad \left\{ \text{displaystyle } S_{n-1}^2 = \frac{1}{n-1} \sum_{i=1}^n \left(X_i - \overline{X} \right)^2 = \frac{1}{n-1} \left(\sum_{i=1}^n X_i^2 - n\overline{X}^2 \right) \right.$$

7.

$$\text{MSE} = \frac{1}{n} \sum_{i=1}^n (e_i)^2 = \frac{1}{n} \mathbf{e}^T \mathbf{e} \quad \left\{ \text{displaystyle } \operatornamename {MSE} = \frac{1}{n} \sum_{i=1}^n (e_i)^2 = \frac{1}{n} \mathbf{e}^T \mathbf{e} \right.$$

8.

$$\text{MSE}(\hat{\theta}) = \text{Var}(\hat{\theta}) + \text{Bias}(\hat{\theta}, \theta)^2 \quad \left\{ \text{displaystyle } \operatornamename {MSE}(\hat{\theta}) = \text{Var}(\hat{\theta}) + \text{Bias}(\hat{\theta}, \theta)^2 \right.$$

9.

Interpretation An MSE of zero, meaning that the estimator $\hat{\theta}$ predicts observations of the parameter θ with perfect accuracy, is ideal (but typically not possible). Values of MSE may be used for comparative purposes. Two or more statistical models may be compared using their MSEs—as a measure of how well they explain a given set of observations: An unbiased estimator (estimated from a statistical model) with the smallest variance among all unbiased estimators is the best unbiased estimator or MVUE (Minimum-Variance Unbiased Estimator). Both analysis of variance and linear regression techniques estimate the MSE as part of the analysis and use the estimated MSE to determine the statistical significance of the factors or predictors under study. The goal of experimental design is to construct experiments in such a way that when the observations are analyzed, the MSE is close to zero relative to the magnitude of at least one of the estimated treatment effects. In one-way analysis of variance, MSE can be calculated by the division of the sum of squared errors and the degree of freedom. Also, the f-value is the ratio of the mean squared treatment and the MSE. MSE is also used in several stepwise regression techniques as part of the determination as to how many predictors from a candidate set to include in a model for a given set of observations.